14. Let G be a group of order pq for p and q primes, $q > p$, and $q \equiv 1 (\text{mod } p)$. By Sylow theory, G contains a normal subgroup H of order q which is cyclic, being of prime order. Let a be a generator of H and let $b \in G, b \notin H$. Now G/H has order p, so $b^p \in H$. If $b^p \neq 1$, then b is of order pq and G is cyclic and thus abelian, so for nonabelian G, we must have $b^p = 1$. Now $bab^{-1} \in H$. If $bab^{-1} = a$, then $ba = ab$ and G is abelian. Thus bab^{-1} must be one of $a^2, a^3, \cdots, a^{q-1}$. By Exercise 13, Part(**b**), the exponents x from 2 to $q-1$ such that the presentation $(a, b : a^q = 1, b^p = 1, ba = a^x b)$ gives a group of order pq are those such that $x^p \equiv 1 (\text{mod } q)$. By Corollary 23.6, the integers $1, 2, 3, \cdots, q-1$ form a cyclic group $\langle \mathbb{Z}_q^*, \cdot \rangle$ of order $q-1$ under multiplication modulo q. Because $q \equiv 1 (\text{mod } p)$ and $p < q$, we see that p divides $q-1$, so there is a cyclic subgroup $\langle r \rangle$ of $\langle \mathbb{Z}_q^*, \cdot \rangle$ having order p, so that $(r^j)^p \equiv 1 \pmod{q}$ for $j = 0, 1, \cdots, p-1$. Thus by Exercise 13, the presentations

$$(a, b : a^q = 1, b^p = 1, ba = a^{(r^j)} b)$$

give groups of order pq for $j = 1, 2, \cdots, p-1$. The hint in the exercise concludes with the demonstration that these $p-1$ presentations are isomorphic.

41. Simplicial Complexes and Homology Groups.

1. **a.** We have $\partial_2(c) = 2\partial_2(P_1P_3P_4) - 4\partial_2(P_3P_4P_6) + 3\partial_2(P_3P_2P_4) + \partial_2(P_1P_6P_4)$
$= 2(P_3P_4 - P_1P_4 + P_1P_3) - 4(P_4P_6 - P_3P_6 + P_3P_4) +$
$3(P_2P_4 - P_3P_4 + P_3P_2) + (P_6P_4 - P_1P_4 + P_1P_6)$
$= 2P_1P_3 - 3P_1P_4 + P_1P_6 - 3P_2P_3 + 3P_2P_4 - 5P_3P_4 + 4P_3P_6 - 5P_4P_6$.

b. No, $\partial_2(c)$ was just computed, and is nonzero.

c. Yes, it is a 1-cycle because $\partial^2 = 0$, that is, $\partial_1(\partial_2(c)) = 0$.

2. We have $\partial_2(\partial_3(P_1P_2P_3P_4)) = \partial_2(P_2P_3P_4 - P_1P_3P_4 + P_1P_2P_4 - P_1P_2P_3)$
$= P_3P_4 - P_2P_4 + P_2P_3 - P_3P_4 + P_1P_4 - P_1P_3 + P_2P_4 - P_1P_4 \quad + P_1P_2 - P_2P_3 + P_1P_3 - P_1P_2$
$= 0$

3. For $i > 0$, all four of these groups are zero. We have $B_0(P) = 0$ while $C_0(P) = Z_0(P) = \{nP \mid n \in \mathbb{Z}\} \simeq \mathbb{Z}$, and $H_0(P) = \{\{nP\} \mid n \in \mathbb{Z}\} \simeq \mathbb{Z}$.

4. For $i > 0$, all four of these groups are zero. We have $B_0(X) = 0$ while $C_0(X) = Z_0(X) = \{mP + nP' \mid m, n \in \mathbb{Z}\} \simeq \mathbb{Z} \times \mathbb{Z}$, and $H_0(X) = \{\{mP + nP'\} \mid m, n \in \mathbb{Z}\} \simeq \mathbb{Z} \times \mathbb{Z}$.

5. For $i > 1$, all four of these groups are zero. We have $C_1(X) = \{nP_1P_2 \mid n \in \mathbb{Z}\} \simeq \mathbb{Z}$. Because $\partial_1(P_1P_2) = -P_1 + P_2 \neq 0$, we have $Z_1(X) = B_1(X) = H_1(X) = 0$. For dimension 0, we have $C_0(X) = Z_0(X) = \{mP_1 + nP_2 \mid m, n \in \mathbb{Z}\} \simeq \mathbb{Z} \times \mathbb{Z}$ and $B_0(X) = \{n(P_2 - P_1) \mid n \in \mathbb{Z}\} \simeq \mathbb{Z}$. Because

$$rP_1 + sP_2 + (-s)(P_2 - P_1) = (r + s)P_1,$$

we see that every coset of $B_0(X)$ in $Z_0(X)$ contains a unique element of the form mP_1 for $m \in \mathbb{Z}$, so $H_0(Z) = Z_0X/B_0(X) = \{mP_1 + B_0(X) \mid m \in \mathbb{Z}\} \simeq \mathbb{Z}$.

6. T F T T T T F T T T

7. a. An **oriented *n*-simplex** $\sigma = P_1P_2\cdots P_{n+1}$ is an ordered sequence of $n+1$ vertices in $\mathbb{R}^m$ where $m \geq n$. If the n-simplex $\mu = P_{k_1}P_{k_2}\cdots P_{k_{n+1}}$ contains the same vertices as σ in a different order, then $\mu = \sigma$ if the permutation

$$\begin{pmatrix} 1 & 2 & \cdots & n+1 \\ k_1 & k_2 & \cdots & k_{n+1} \end{pmatrix}$$

is even and $\mu = -\sigma$ if the permutation is odd.

b. We have

$$\partial_n(P_1P_2\cdots P_{n+1}) = \sum_{i=1}^{n+1}(-1)^{i+1}P_1P_2\cdots P_{i-1}P_{i+1}\cdots P_{n+1}.$$

c. Each summand of the boundary of an n-simplex is a **face** of the simplex.

8. They are already defined in the text in terms of a general integer $n \geq 0$. Take them just as they stand.

9. Let $\sigma = P_1P_2\cdots P_{n+1}$. The ith face of $\partial_n(\sigma)$ is

$$(-1)^{i+1}P_1\cdots P_{i-1}P_{i+1}\cdots P_{n+1}$$

and the jth face is $(-1)^{j+1}P_1\cdots P_{j-1}P_{j+1}\cdots P_{n+1}$. Let us suppose that $i < j$. Applying ∂_{n-1} to the ith face produces an (n-2)-chain containing the term

$$m_iP_1\cdots P_{i-1}P_{i+1}\cdots P_{j-1}P_{j+1}\cdots P_{n+1}$$

where $m_i = (-1)^{i+1}(-1)^j$, for the vertex P_j became the $(j-1)$st vertex in the ith face after its predecessor P_i was removed. On the other hand, applying ∂_{n-1} to the jth face produces this same $(n-2)$-simplex with coefficient $m_j = (-1)^{j+1}(-1)^{i+1}$, for P_i was still the ith vertex in the jth face since $i < j$. Thus these two terms cancel each other in the computation of $\partial_{n-1}(\partial_n(\sigma))$.

10. a. Because P_1 is a *summand* of ∂_1 of P_2P_1, P_3P_1, and P_4P_1, we have $\delta^{(0)}P_1 = P_2P_1 + P_3P_1 + P_4P_1$. Similarly, $\delta^{(0)}(P_4) = P_1P_4 + P_2P_4 + P_3P_4$.

b. Note that P_3P_2 is a *summand* of both ∂_2 of $P_1P_3P_2$ and ∂_2 of $P_3P_2P_1$. However, as stated in the text, these 2-simplexes are the same because we consider $P_1P_3P_2 = P_3P_2P_1 = P_2P_1P_3$. Thus $\delta^{(1)}(P_3P_2) = P_1P_3P_2 + P_4P_3P_2$.

c. We have $\delta^{(2)}(P_3P_2P_4) = P_1P_3P_2P_4$.

11. a. We define $\delta^{(n)} : C^{(n)} \to C^{(n+1)}$ by

$$\delta^{(n)}\left(\sum_i m_i\sigma_i\right) = \sum_i m_i\,\delta^{(n)}(\sigma_i).$$

b. It suffices to show that $\delta^{(n+1)}(\delta^{(n)}(\sigma)) = 0$ for every n-simplex σ. In the case where σ is not a face of a face of an (n+2)-simplex, the conclusion is obvious. Otherwise, let σ be a face of a face of an (n+2)-simplex that contains two additional vertices, P_i and P_j. Now $\delta^{(n)}(\sigma)$ contains both $P_i\sigma$ and $P_j\sigma$ as *summands*. Then $\delta^{(n+1)}(\delta^{(n)}(\sigma))$ contains both $P_jP_i\sigma$ and $P_iP_j\sigma$ as *summands*. Because the reordering of the vertices of $P_jP_i\sigma$ to produce $P_iP_j\sigma$ is accomplished by an odd permutation, namely the single transposition (i, j), we see that the second simplex is the negative the the first, so these two terms cancel each other in $C^{(n+2)}(X)$. Thus $\delta^2 = 0$.